PROJECT PORTFOLIO

Proposed 4-Storied Residential Building

Architectural and structural drawing set

24 drawings | Architectural, structural, utilities and location

OWNER	Mst. Anjuman Ara Begum, D/O Kabibor Rahman
SITE	Mouza Bagulagarhi, Kishoreganj, Nilphamari, Bangladesh
PLOT REFERENCE	Khatiyan No. 1079, Dag No. 452, J.L. No. 23
TOTAL LAND AREA	9.00 decimal
GROUND FLOOR AREA	2,782.85 sft
TOTAL FLOOR AREA	11,131.40 sft (as stated on the drawing sheet)
CONSULTANT	Landmark Engineering Solutions, Khoda Hafej Gate, Parbatipur Road, Saidpur, Nilphamari
PREPARED BY	Engr. Mobarok Hossain, B.Sc. Engg (WUB)
CHECKED BY	Engr. Saeed Sourav, B.Sc. Engg (Civil)

Index of Drawings

A. Site & Architectural

01	Site Plan	3
02	Proposed Site Layout Plan	4
03	Ground Floor Plan	5
04	Typical Floor Plan	6
05	Roof Plan	7
06	Front Side Elevation	8
07	Back Side Elevation	9
08	Cross Section (A-A)	10

B. Structural Layouts

09	Grid Layout Plan	11
10	Column Layout Plan	12
11	Footing Layout Plan	13
12	Grade Beam Layout Plan	14
13	Floor Beam Layout Plan (Typical)	15
14	Ground Floor Slab Details	16
15	Floor Beam Layout Plan with Stair Opening	17

C. Structural Details

16	Column Schedule	18
17	Typical Footing Section X-X	19
18	Footing Schedule and Typical Footing Plan	20
19	Long Sections of Grade Beams GB1, GB1A and GB2	21
20	Long Sections of Grade Beams GB2A and GB3, Typical Section	22
21	Stair Details	23

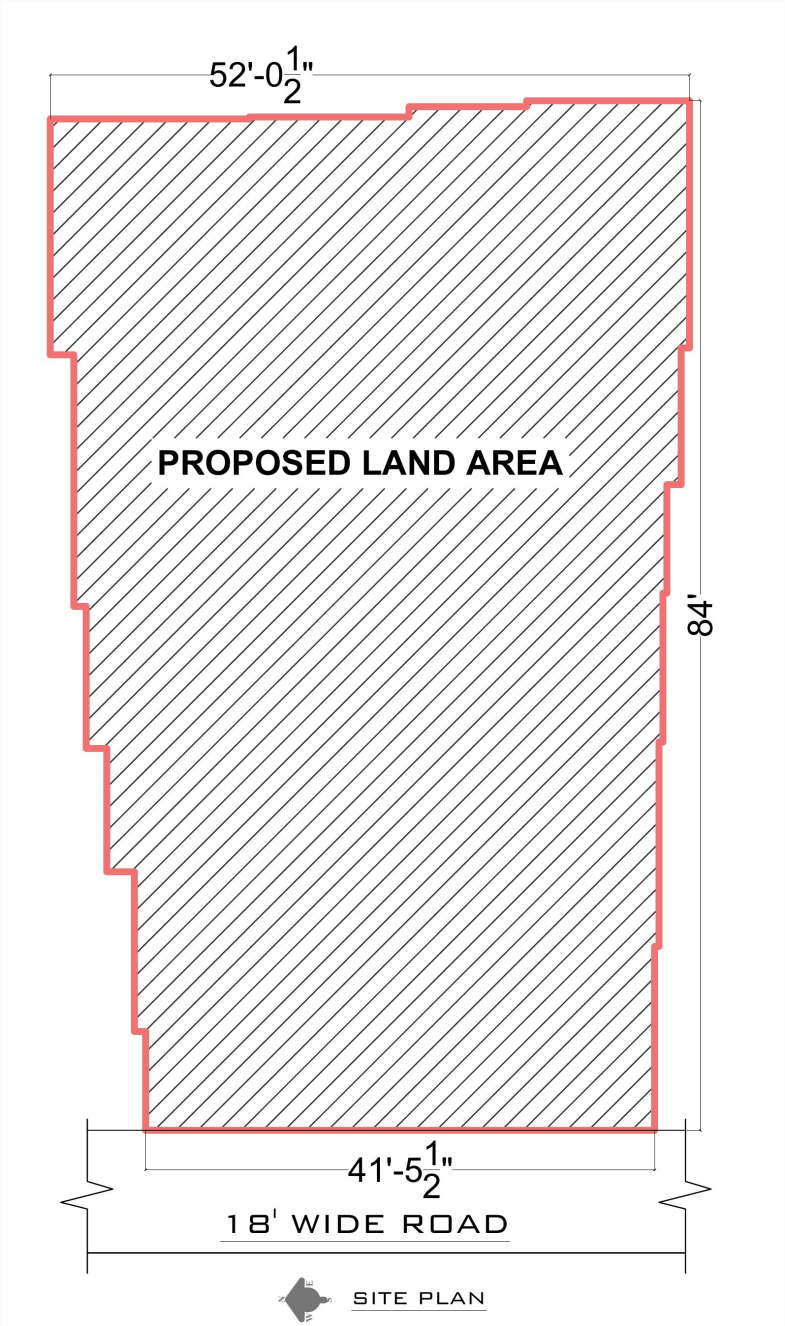
D. Utilities

22	Septic Tank, Soak Well and Underground Reservoir Details	24

E. Location

23	Mouza Map	25
24	Location Map	26

Site Plan



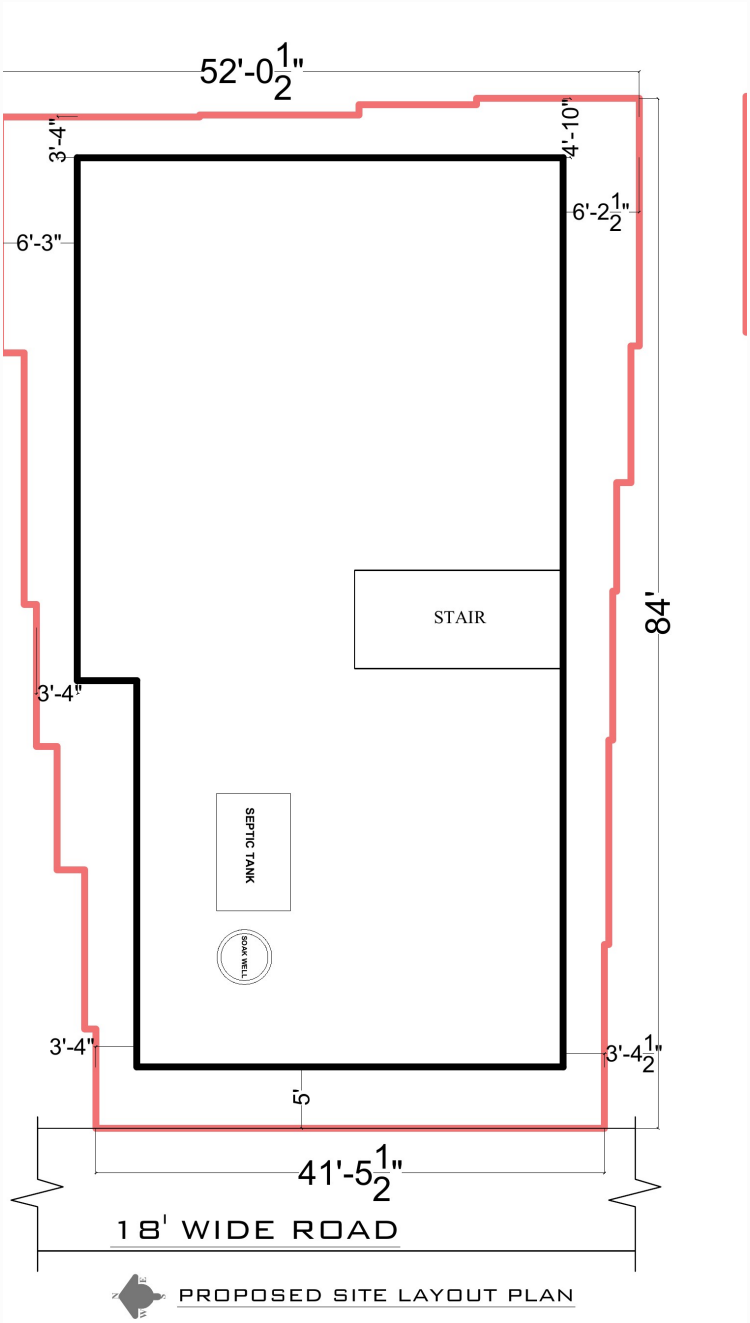
DESCRIPTION

Shows the proposed land area (hatched) as an irregular plot facing an 18' wide road on its south side. The plot is the basis for all later layouts and for the setback and coverage checks.

KEY INFORMATION FROM THE DRAWING

- Plot width 52'-0½" at the rear and 41'-5½" at the road frontage
- Plot depth 84'-0"
- Total land area 9.00 decimal (Khatiyan 1079, Dag 452, J.L. No. 23)

Proposed Site Layout Plan



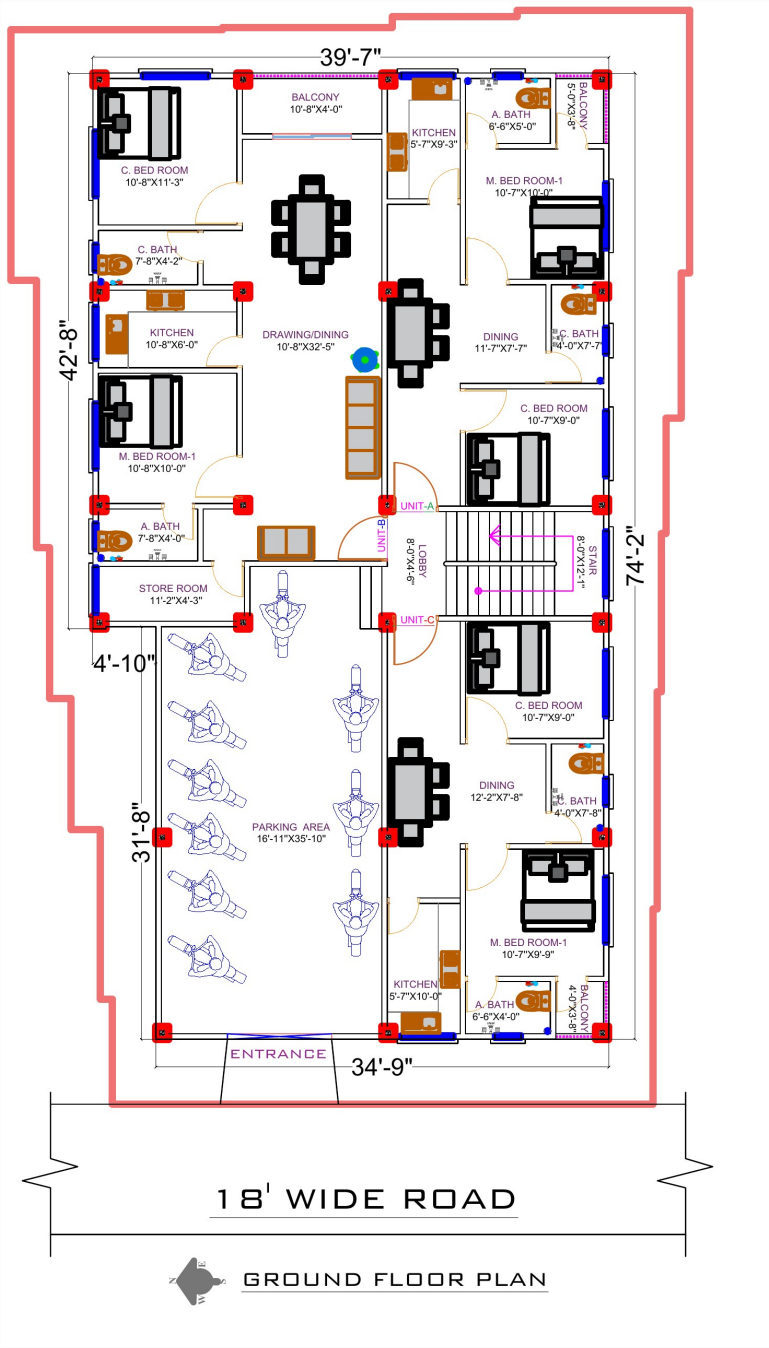
DESCRIPTION

Places the building footprint on the plot and locates the services around it. It shows the stair position and the septic tank and soak well, with the open spaces left between the building and the plot boundary.

KEY INFORMATION FROM THE DRAWING

- Building footprint outlined in bold within the plot boundary
- Setback dimensions marked between 3'-4" and 6'-3" around the footprint
- Septic tank and soak well located in the rear open area
- Frontage on the 18' wide road

Ground Floor Plan



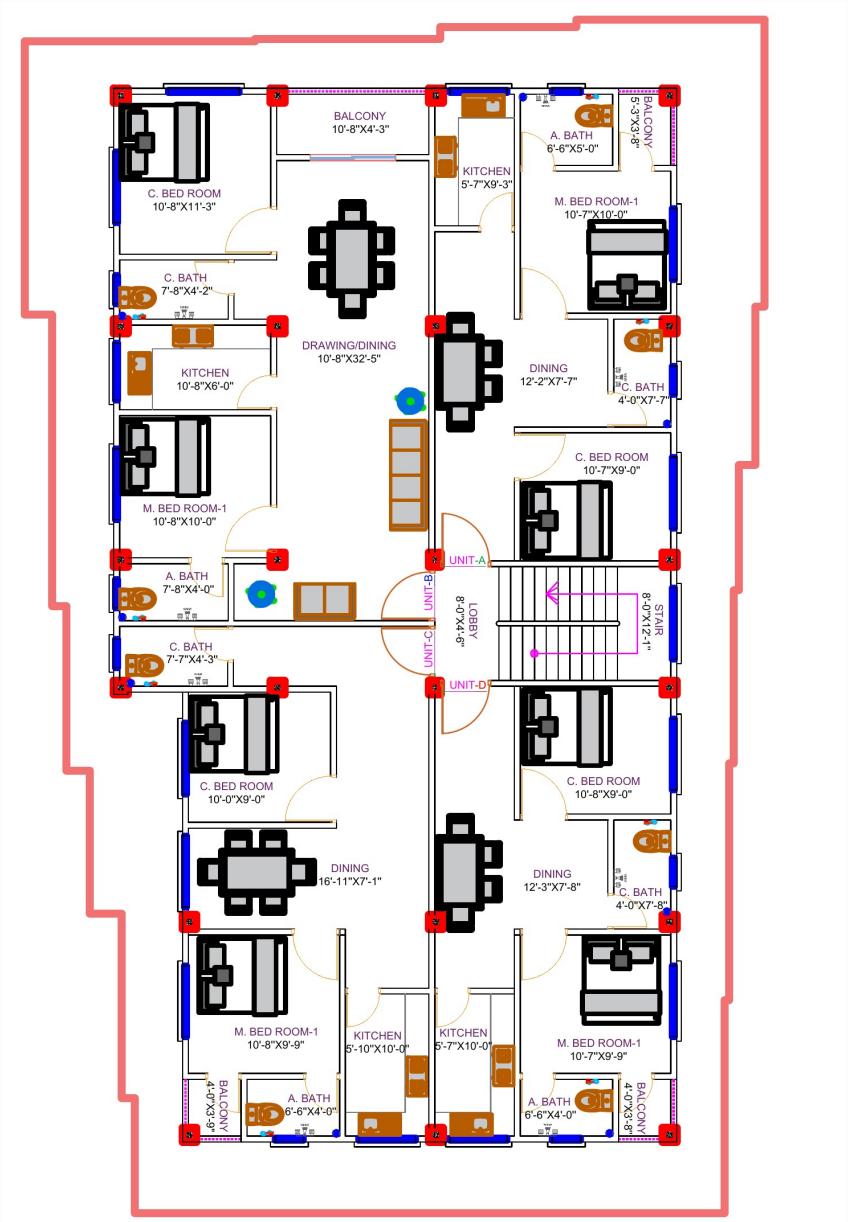
DESCRIPTION

Furnished layout of the ground floor, showing room arrangement, door and window openings, stairs and the entrance from the road. Besides the residential spaces it includes the parking area and a store room.

KEY INFORMATION FROM THE DRAWING

- Parking area 16'-11" x 35'-10"
- Store room 11'-2" x 4'-3"
- Bedrooms about 10'-7" x 10'-0" with attached baths, kitchens and dining spaces
- Overall width dimension 38'-7" shown across the plan

Typical Floor Plan



TYPICAL FLOOR PLAN

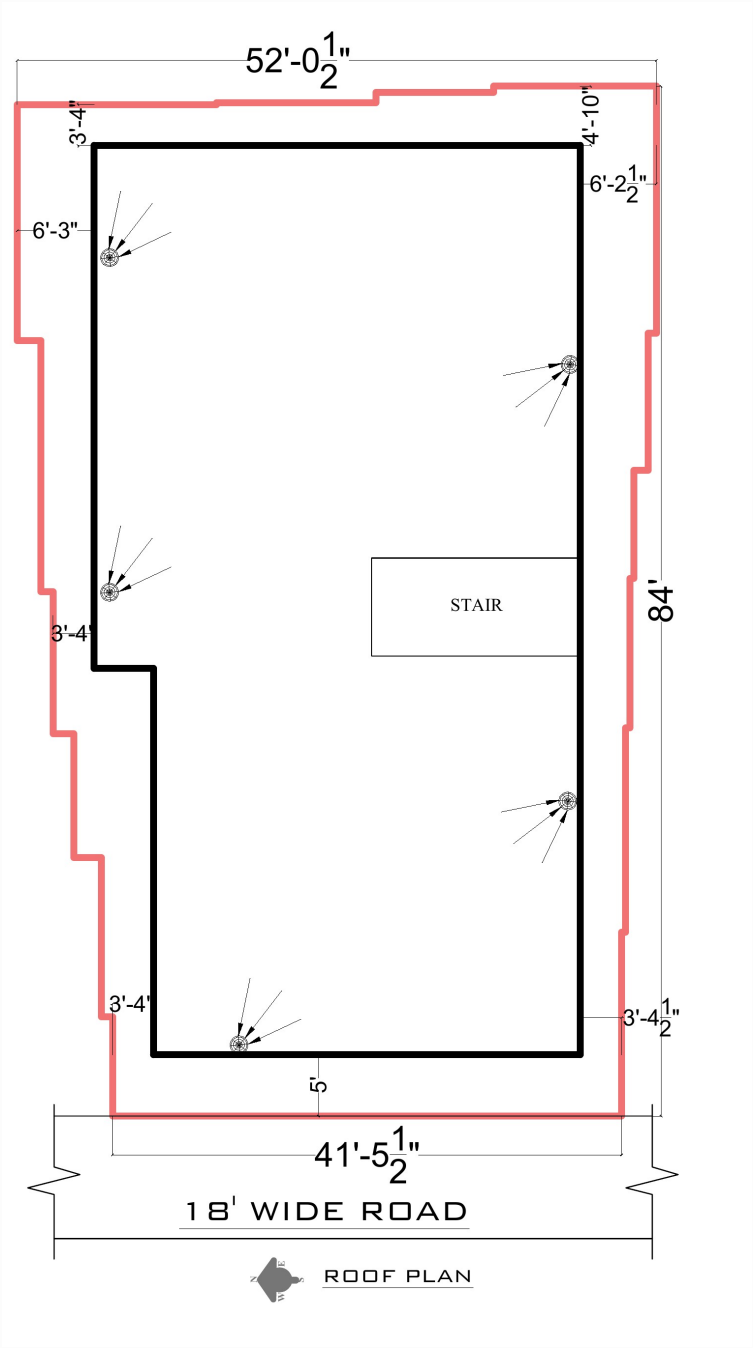
DESCRIPTION

Furnished plan of the repeating residential floor used for the 1st to 3rd floors. Each floor holds several apartment units, each with bedrooms, attached bath, kitchen, dining space and balcony, served by a common stair and lobby.

KEY INFORMATION FROM THE DRAWING

- Master bedrooms about 10'-7" x 10'-0", child bedrooms about 10'-7" x 9'-0"
- Kitchens from 5'-7" x 9'-3" to 10'-8" x 6'-0"
- Balconies and attached baths for every unit
- Drawing/dining space 10'-8" x 32'-5"

Roof Plan



DESCRIPTION

Plan view of the roof slab at the top of the building, including the stair head-room outline, on the same plot outline as the site layout.

KEY INFORMATION FROM THE DRAWING

- Same footprint and setback dimensions as the layout plan
- Stair head-room position marked
- Plot 52'-0½" x 84'-0" with 18' wide road to the south

Front Side Elevation



DESCRIPTION

Elevation of the road-facing facade, showing the four storeys, window openings, balcony slabs and the ground floor entrance and parking opening.

KEY INFORMATION FROM THE DRAWING

- Four storeys (ground + 3 upper floors)
- Window and balcony layout repeated on each floor

Back Side Elevation



BACK SIDE ELEVATION

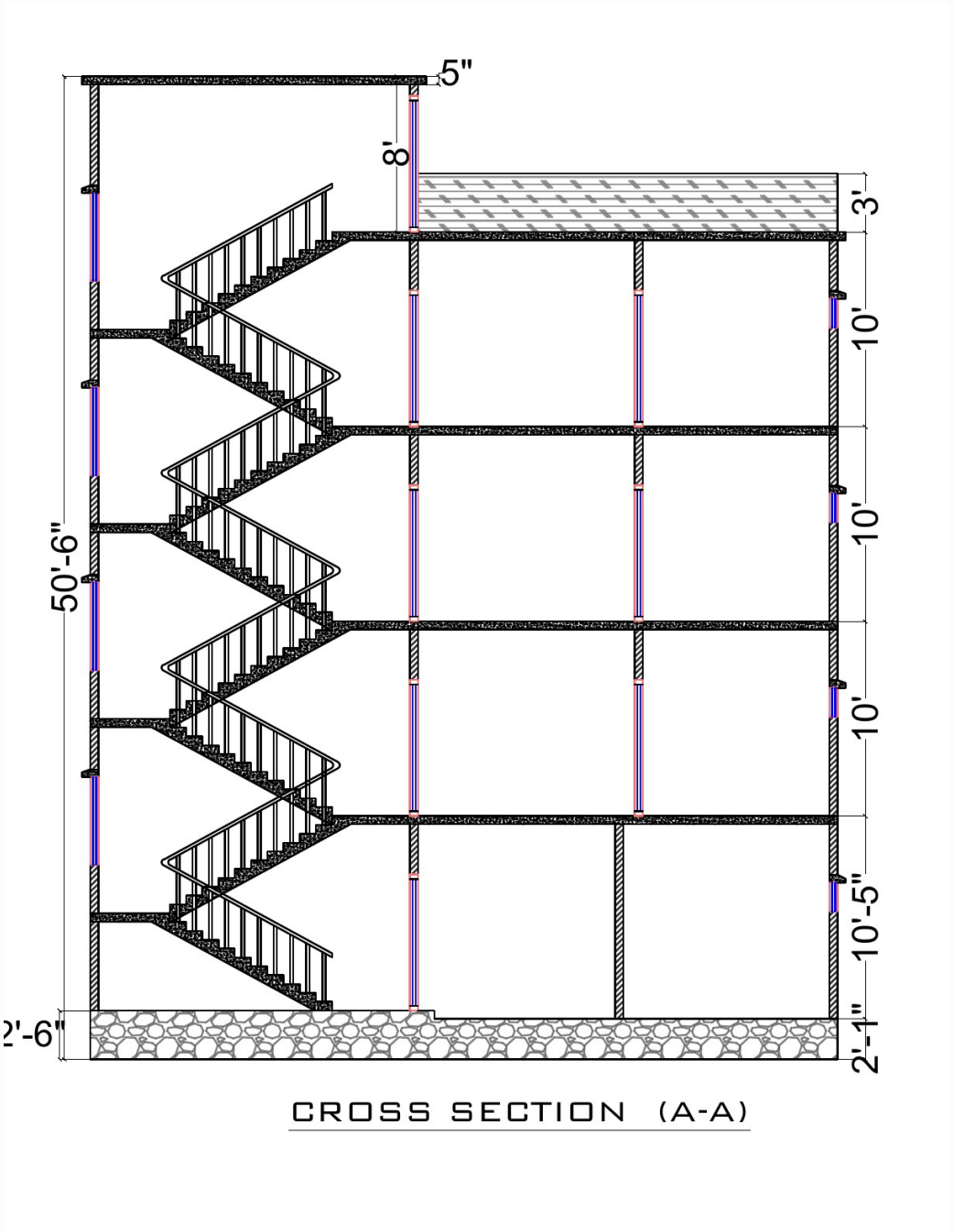
DESCRIPTION

Elevation of the rear facade, showing the four storeys with windows and balconies and the planter greenery drawn along the balconies.

KEY INFORMATION FROM THE DRAWING

- Four storeys, same height as the front elevation
- Balcony and window arrangement for the rear units

Cross Section (A-A)



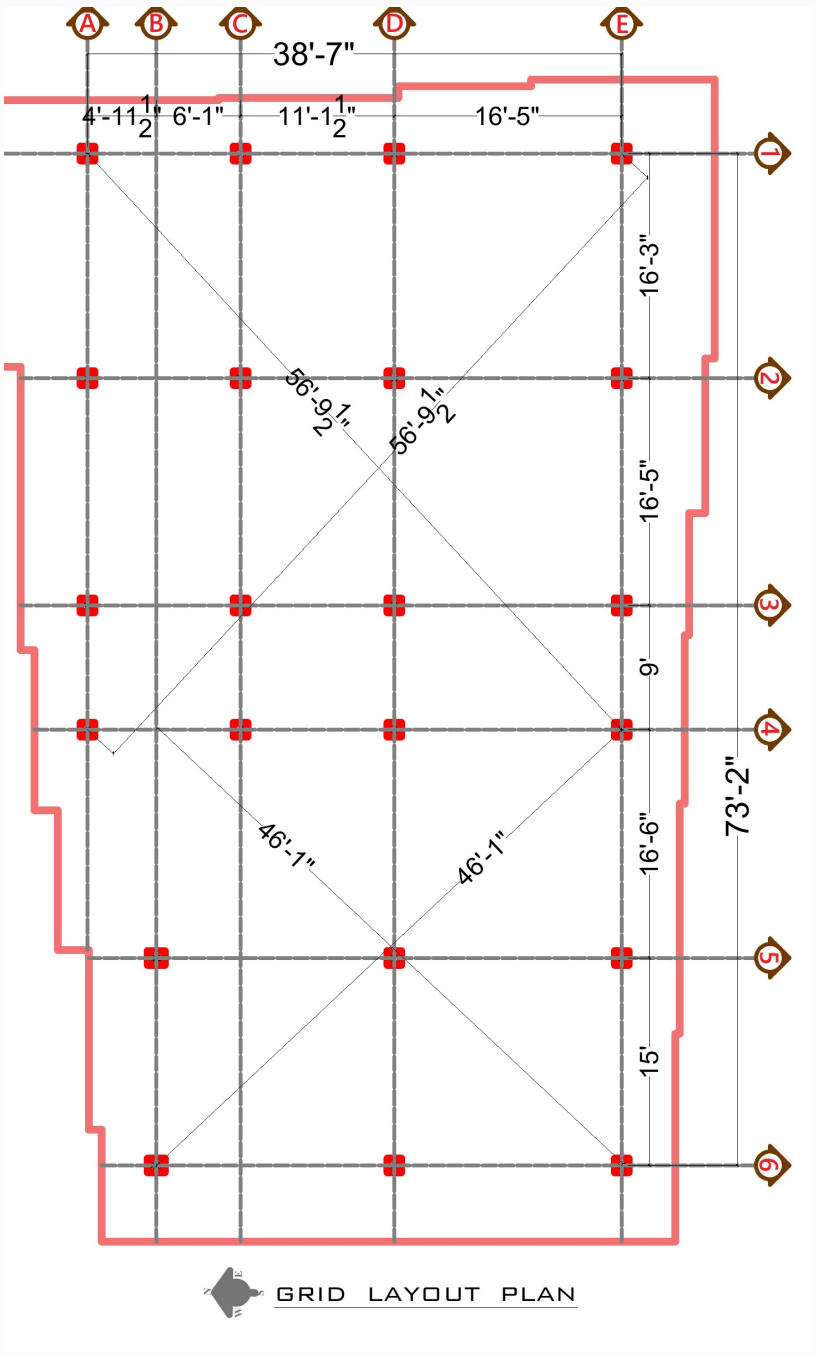
DESCRIPTION

Vertical section through the building along the stair. It shows the floor-to-floor heights, the flights and landings of the stair, and the foundation and ground levels.

KEY INFORMATION FROM THE DRAWING

- Floor heights of 10'-0" per storey as dimensioned
- Overall height dimension 50'-6"
- Stair flights and landings shown on each level

Grid Layout Plan



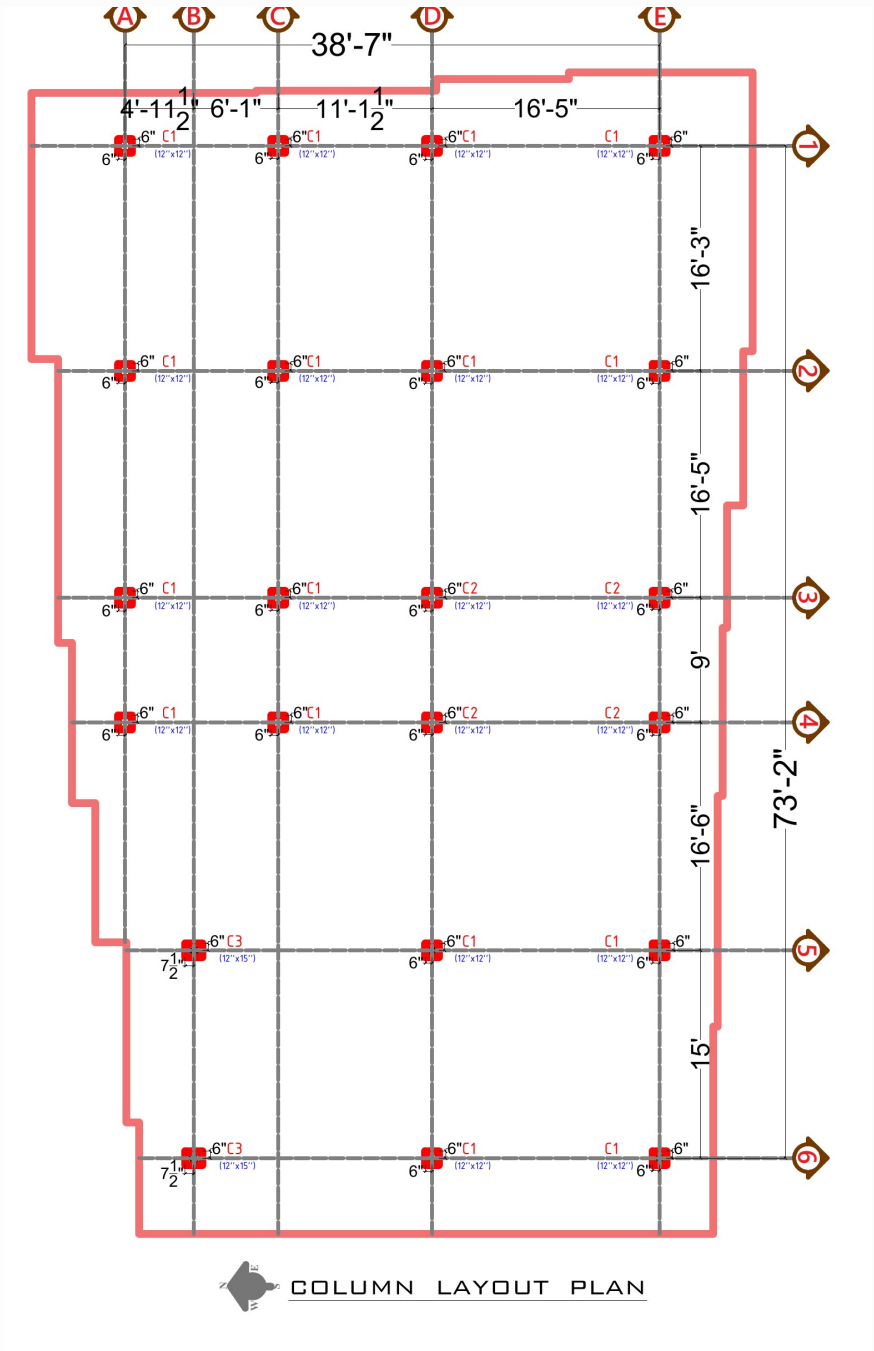
DESCRIPTION

Column grid used to set out the structure. Gridlines A to E run in one direction and 1 to 6 in the other, with spacing dimensions marked between them.

KEY INFORMATION FROM THE DRAWING

- Grid lines A-E and 1-6
- Overall length 73'-2"
- Bay dimensions include 16'-5", 16'-3", 16'-6", 15' and 38'-7"

Column Layout Plan



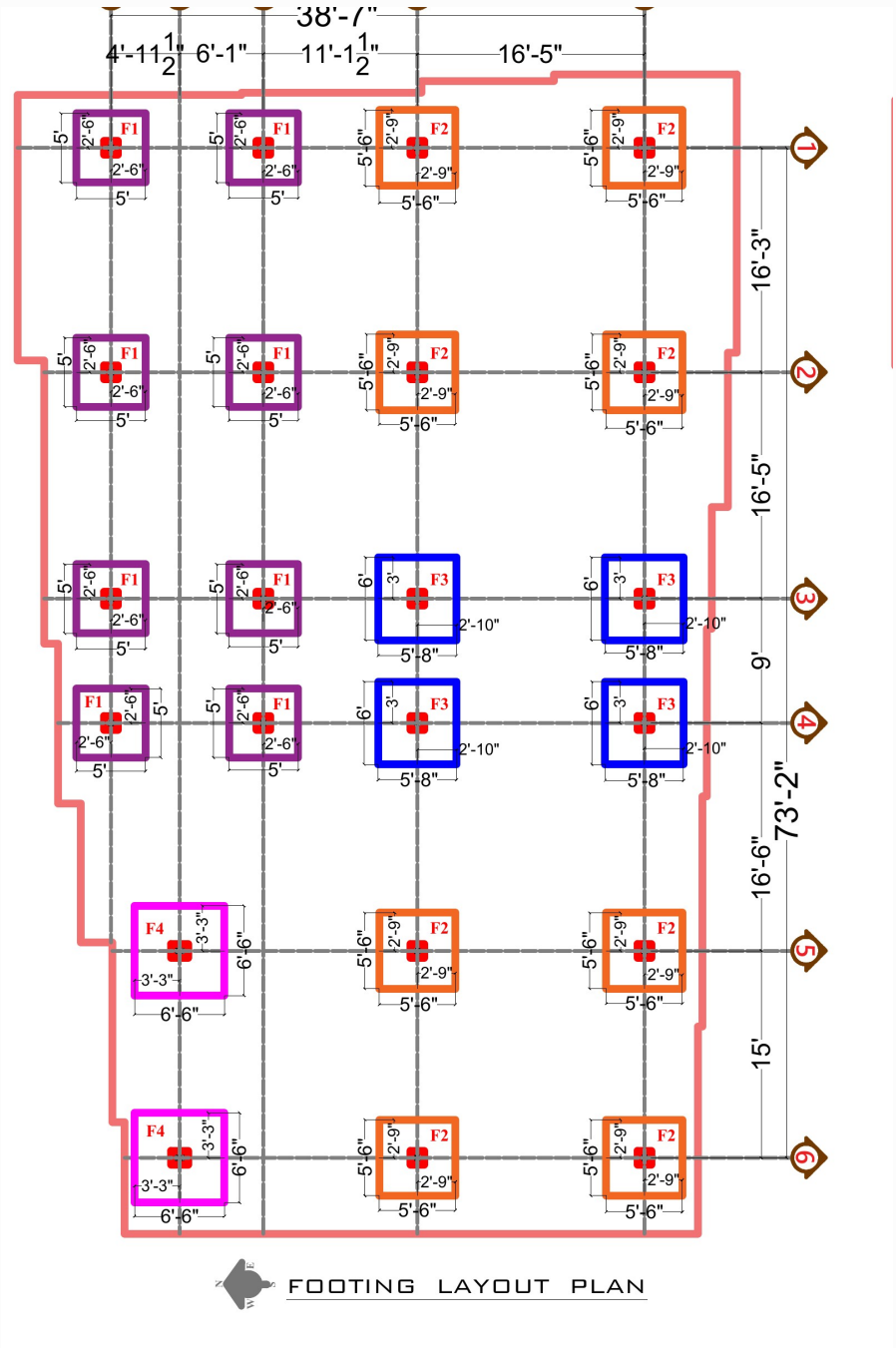
DESCRIPTION

Locations and marks of all columns on the grid. Three column types are used, and each is drawn with its size and cover dimension.

KEY INFORMATION FROM THE DRAWING

- C1 and C2: 12" x 12" at plan level
- C3: 12" x 15"
- Schedule of reinforcement is on the Column Schedule sheet

Footing Layout Plan



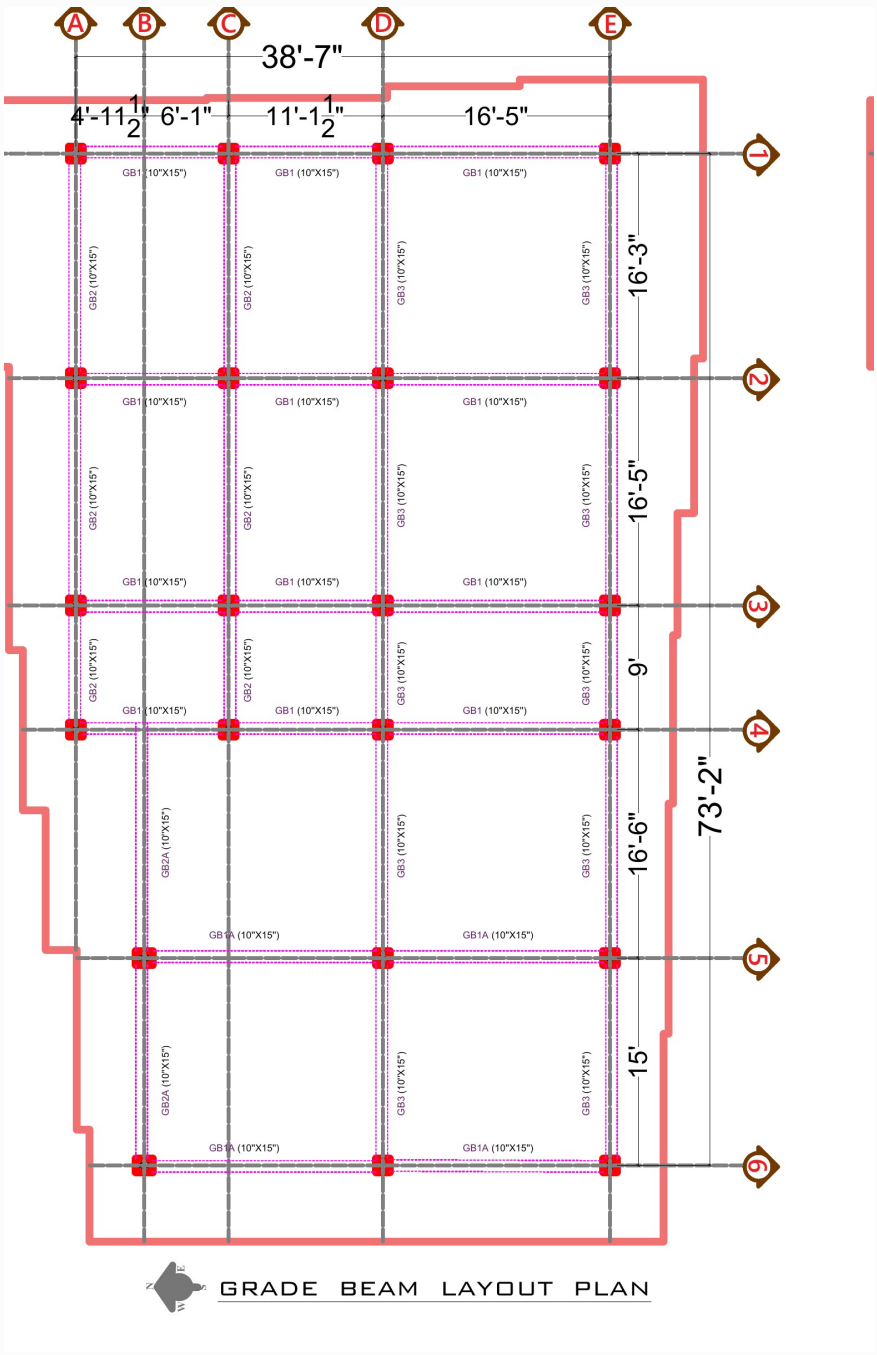
DESCRIPTION

Isolated footings below each column, marked F1 to F4 according to column load. Footing sizes are dimensioned on the plan and tabulated in the Footing Schedule.

KEY INFORMATION FROM THE DRAWING

- F1 5'-0" x 5'-0", 14" thick
- F2 5'-6" x 5'-6", 15" thick
- F3 6'-0" x 5'-8", 16" thick
- F4 6'-6" x 6'-6", 17" thick

Grade Beam Layout Plan



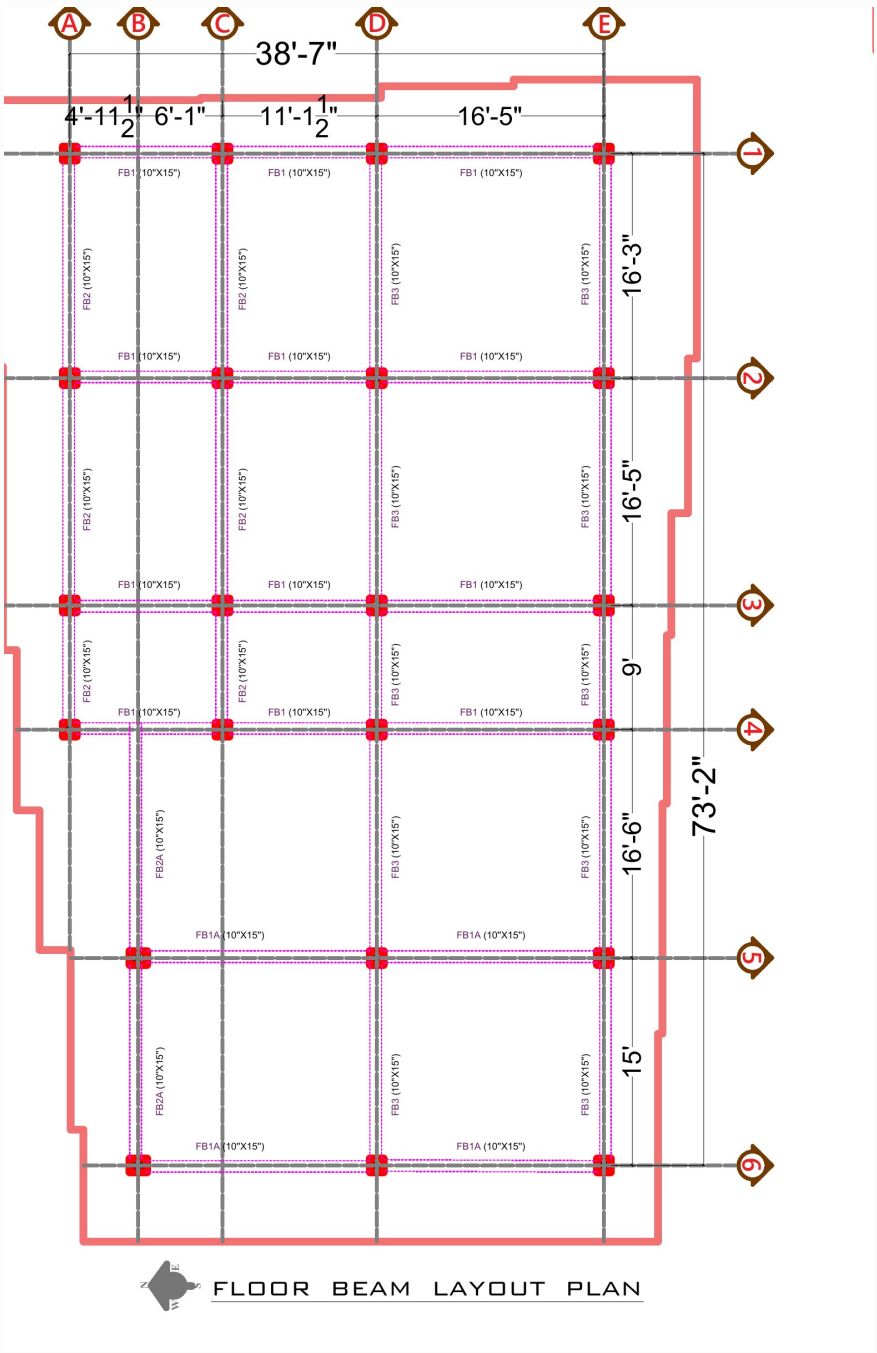
DESCRIPTION

Plan of the grade (plinth-level) beams that tie the footings and columns together. Each beam carries a mark and size.

KEY INFORMATION FROM THE DRAWING

- Beam marks GB1, GB1A, GB2, GB2A and GB3
- All beams 10" x 15"
- Reinforcement shown on the grade beam long sections

Floor Beam Layout Plan (Typical)



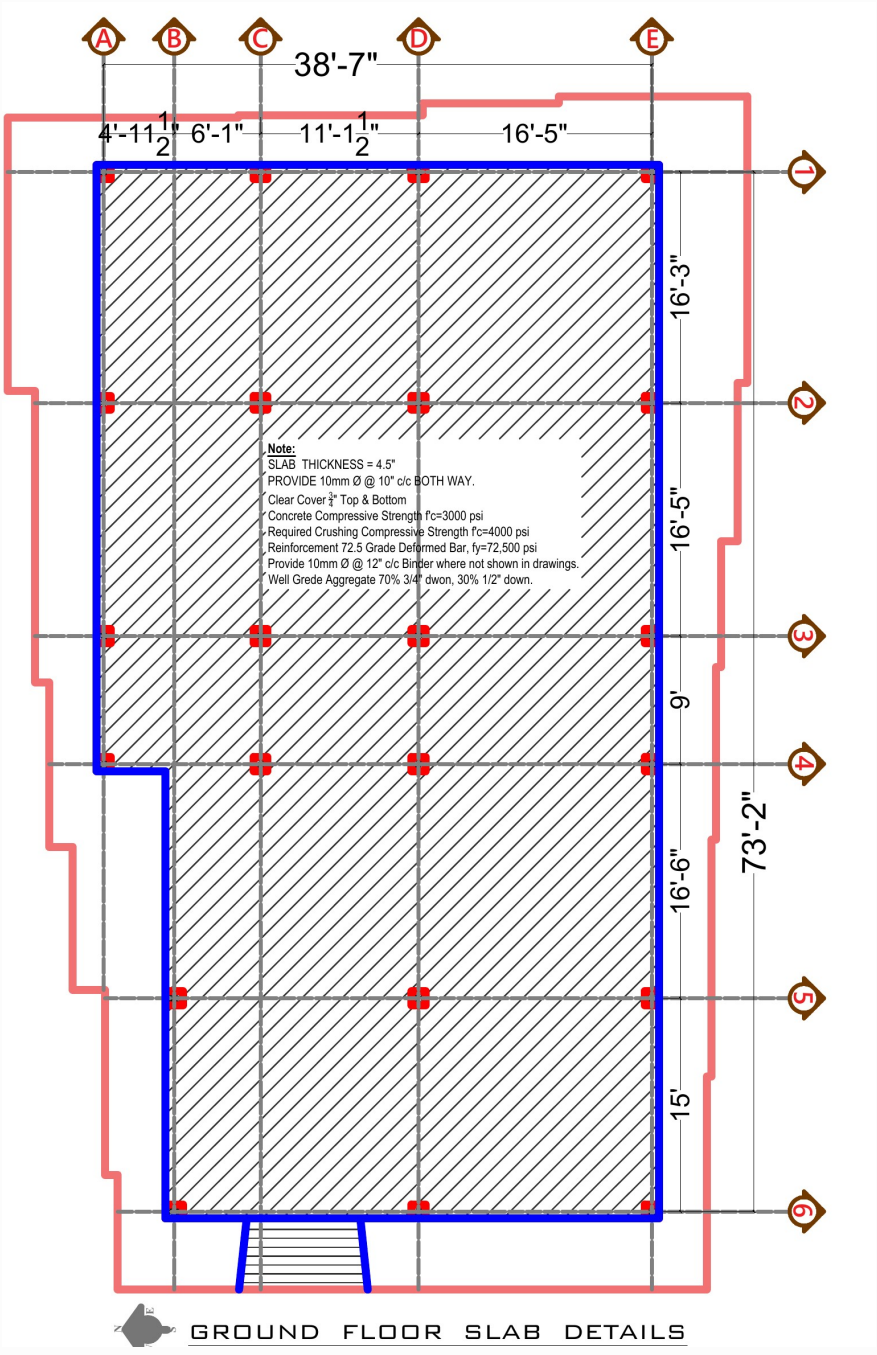
DESCRIPTION

Layout of the floor-level beams of the typical floor, each marked with its type and size.

KEY INFORMATION FROM THE DRAWING

- Beam marks FB1, FB1A, FB2, FB2A and FB3
- All beams 10" x 15"
- Same grid A-E and 1-6 with 73'-2" overall length

Ground Floor Slab Details



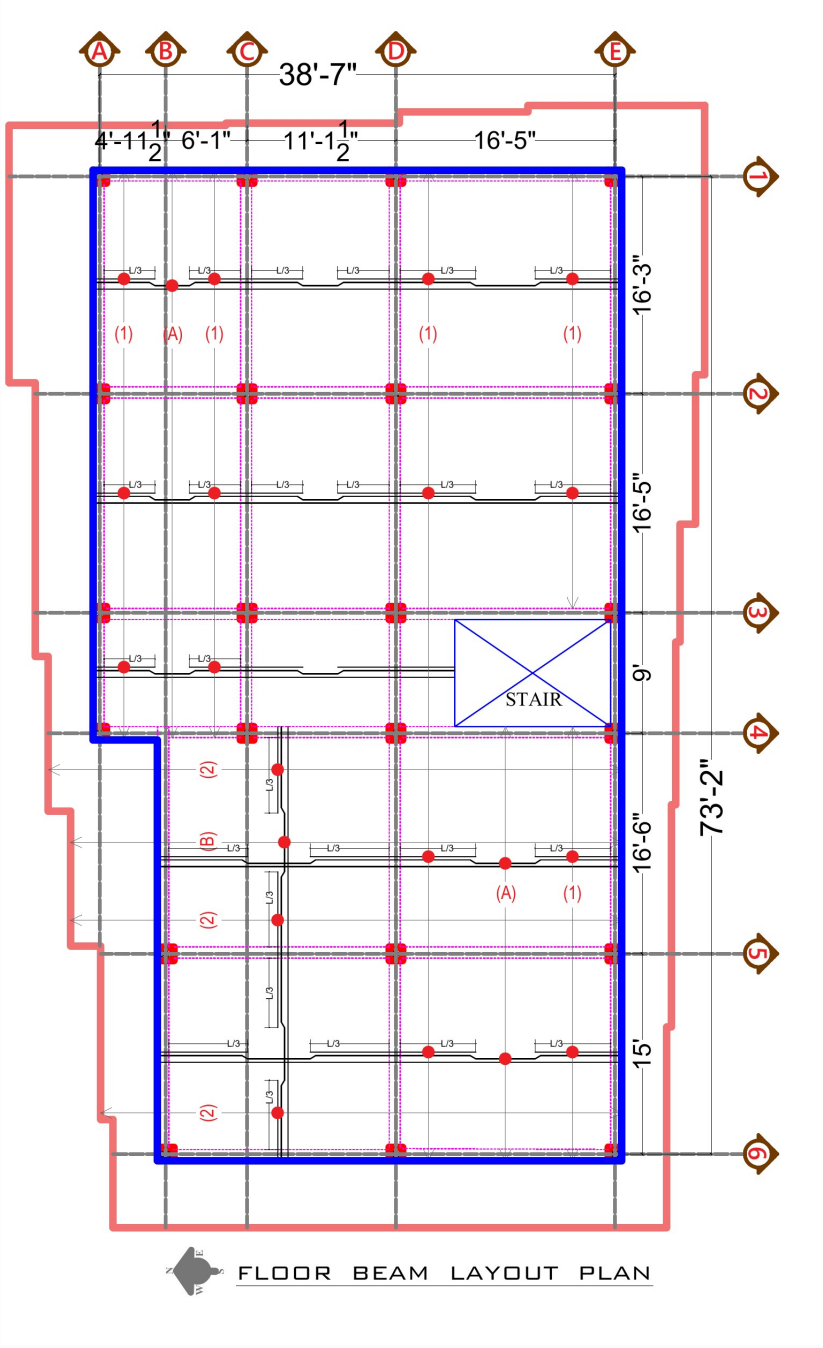
DESCRIPTION

Plan of the ground floor slab with the general notes for slab thickness, reinforcement and materials.

KEY INFORMATION FROM THE DRAWING

- Slab thickness 4.5"
- 10 mm dia bars @ 10" c/c both ways
- Clear cover 3/4" top and bottom
- Concrete f'c 3000 psi; required average strength 4000 psi
- Reinforcement 72.5 grade deformed bars, fy 72,500 psi

Floor Beam Layout Plan with Stair Opening



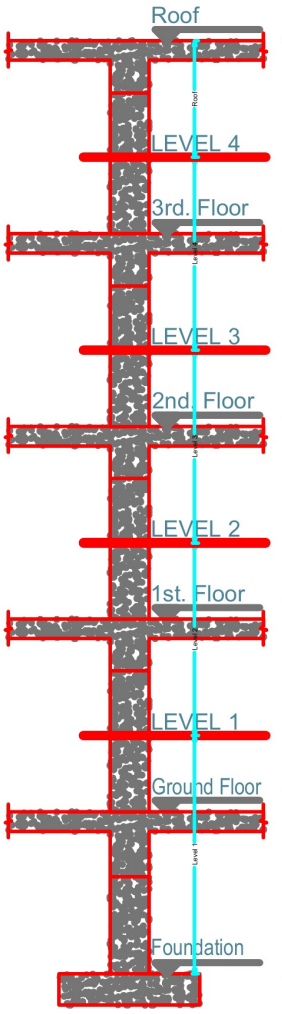
DESCRIPTION

Second floor beam plan, drawn with the stair opening and the L/3 zone marks used for placing reinforcement along the beams.

KEY INFORMATION FROM THE DRAWING

- Stair opening and stair beam positions shown
- L/3 zone marks along beam lines
- Beam and slab panel numbering for placing bars

Column Schedule



LEVEL	C1	C2	C3
Level 4-Roof	6-16mmØ Covering 1.5" 10mmØ @ 5"≈7"≈5" c/c. 12"X12"	8-16mmØ Covering 1.5" 10mmØ @ 5"≈7"≈5" c/c. 12"X12"	10-16mmØ Covering 1.5" 10mmØ @ 5"≈7"≈5" c/c. 12"X15"
Level 3-4	6-16mmØ Covering 1.5" 10mmØ @ 5"≈7"≈5" c/c. 12"X12"	8-16mmØ Covering 1.5" 10mmØ @ 5"≈7"≈5" c/c. 12"X12"	10-16mmØ Covering 1.5" 10mmØ @ 5"≈7"≈5" c/c. 12"X15"
Level 2-3	6-16mmØ Covering 1.5" 10mmØ @ 5"≈7"≈5" c/c. 12"X12"	8-16mmØ Covering 1.5" 10mmØ @ 5"≈7"≈5" c/c. 12"X12"	10-16mmØ Covering 1.5" 10mmØ @ 5"≈7"≈5" c/c. 12"X15"
Level 1-2	6-16mmØ Covering 1.5" 10mmØ @ 5"≈7"≈5" c/c. 12"X12"	8-16mmØ Covering 1.5" 10mmØ @ 5"≈7"≈5" c/c. 12"X12"	10-16mmØ Covering 1.5" 10mmØ @ 5"≈7"≈5" c/c. 12"X15"
Upto Level 1	6-16mmØ Covering 2.5" 10mmØ @ 5" c/c. 14"X14"	8-16mmØ Covering 2.5" 10mmØ @ 5" c/c. 14"X14"	10-16mmØ Covering 2.5" 10mmØ @ 5" c/c. 14"X17"

COLUMN SCHEDULE

Notes for Column:

1. Concrete design strength, f'c = 3500 psi.
2. Required average strength, f'c = 4500 psi.
3. All Reinforcement bars should be deformed bar having yield strength, fy = 72,500 psi.(72.5 Grade)
4. Concrete mix ratio (1:1.5:3) confirm by trial mix ratio
5. If any information is not given should be taken from General Notes.
6. Well Graded Course Aggregate(70% 3/4" Down, 30% 1/2" Down).

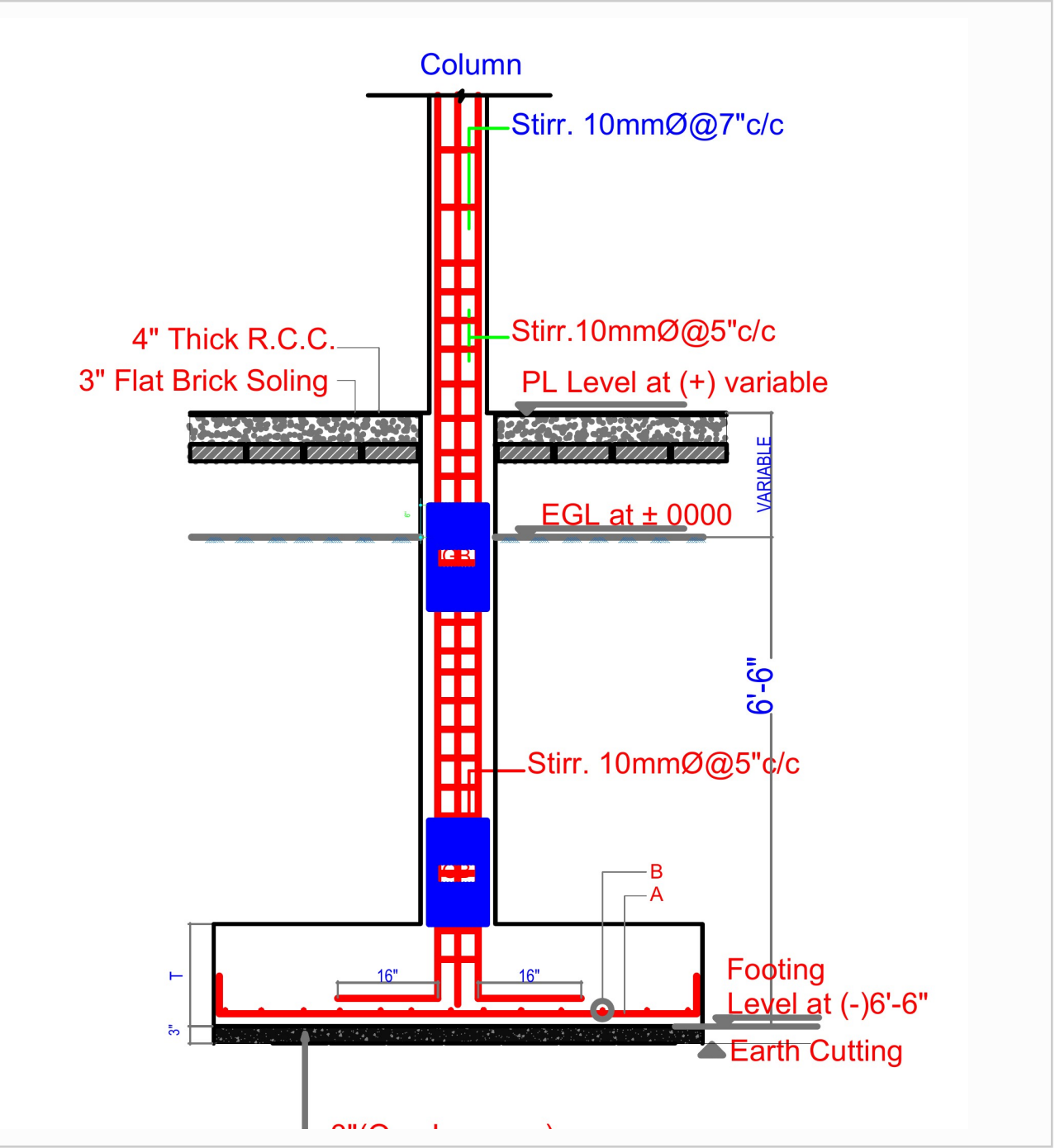
DESCRIPTION

Table of the column types by level, from the foundation up to the roof. For each level and type it gives the section size, the longitudinal bars, the tie bars and the cover.

KEY INFORMATION FROM THE DRAWING

- C1 and C2 section sizes range from 12" x 12" to 14" x 14"; C3 from 12" x 15" to 14" x 17"
- Longitudinal bars 6, 8 or 10 nos of 16 mm dia
- Ties 10 mm dia @ 5" to 7" c/c
- Concrete f'c 3500 psi, required average 4500 psi, mix 1:1.5:3
- Reinforcement 72.5 grade, fy 72,500 psi

Typical Footing Section X-X



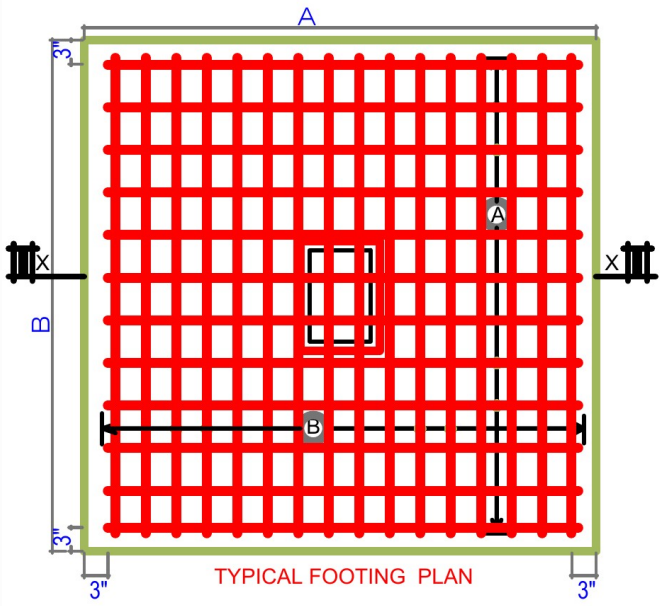
DESCRIPTION

Cross-section through a typical footing, showing the column starter bars and ties, the footing slab, the brick soling and the levels of the footing and grade beam.

KEY INFORMATION FROM THE DRAWING

- Footing level (-)6'-6" below EGL (±0.000)
- 3" flat brick soling and 4" thick R.C.C. layer
- Column ties 10 mm dia @ 5" and @ 7" c/c
- Clear cover 3" to the footing bars

Footing Schedule and Typical Footing Plan



FOOTING SCHEDULE :

Sl. No.	L (Ft.)	B (Ft.)	T	A Mainbar	B Binder
F_1	5'-0"	5'-0"	14"	16 mmØ@ 6"c/c	16 mmØ@ 6"c/c
F_2	5'-6"	5'-6"	15"	16 mmØ@ 6"c/c	16 mmØ@ 6"c/c
F_3	6'-0"	5'-8"	16"	16 mmØ@ 6"c/c	16 mmØ@ 6"c/c
F_4	6'-6"	6'-6"	17"	16 mmØ@ 6"c/c	16 mmØ@ 6"c/c

Note :
1). Clear Cover of the Footing will be 3" in all sides
2). Concrete design strength, f'c = 3000 psi.
3). Required average strength, f'c = 4000 psi.
4). All Reinforcement bars should be deformed bar having yield strength, fy = 72,500 psi.(72.5 Grade)
5). Concrete mix ratio (1:1.5:3) confirm by trial mix ratio
6). If any information is not given should be taken from General Notes.
8). Well Graded Course Aggregate(70% 3/4" Down, 30% 1/2" Down).
9). A = Perpendicular Bar To Long Side Of Column (Lower Layer)
10). B = Perpendicular Bar To Short Side Of Column (Upper Layer)

FOOTING SCHEDULE

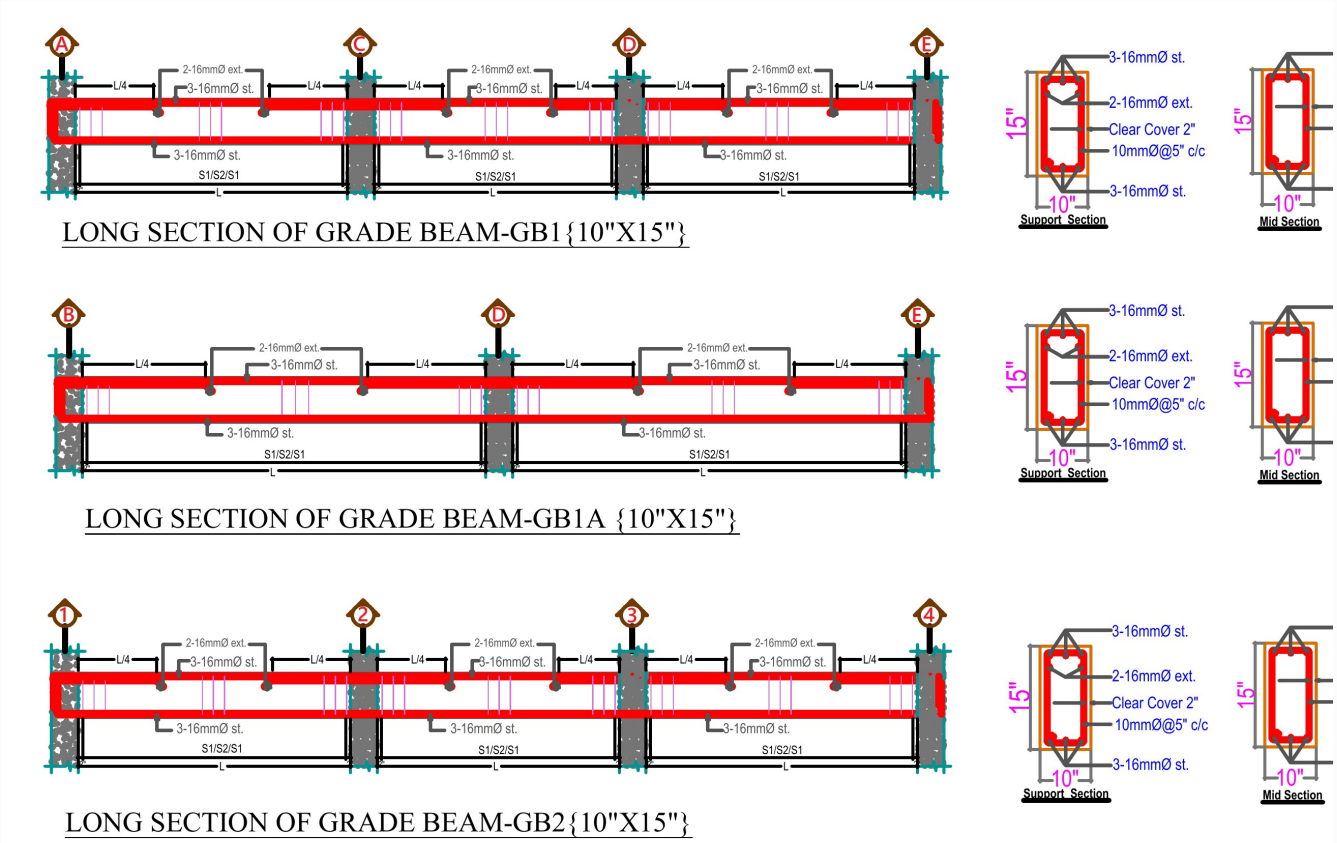
DESCRIPTION

Typical footing plan with the bar mesh, together with the footing schedule table listing the length, breadth, depth and bar spacing of each footing type and the notes that apply to all.

KEY INFORMATION FROM THE DRAWING

- F1 to F4 footing sizes and depths 14" to 17"
- Bars 16 mm dia @ 6" c/c in both layers
- Clear cover 3" on all sides
- Concrete f'c 3000 psi, required average 4000 psi; fy 72,500 psi

Long Sections of Grade Beams GB1, GB1A and GB2



LONG SECTION OF GRADE BEAM-GB1{10"X15"}

LONG SECTION OF GRADE BEAM-GB1A {10"X15"}

LONG SECTION OF GRADE BEAM-GB2{10"X15"}

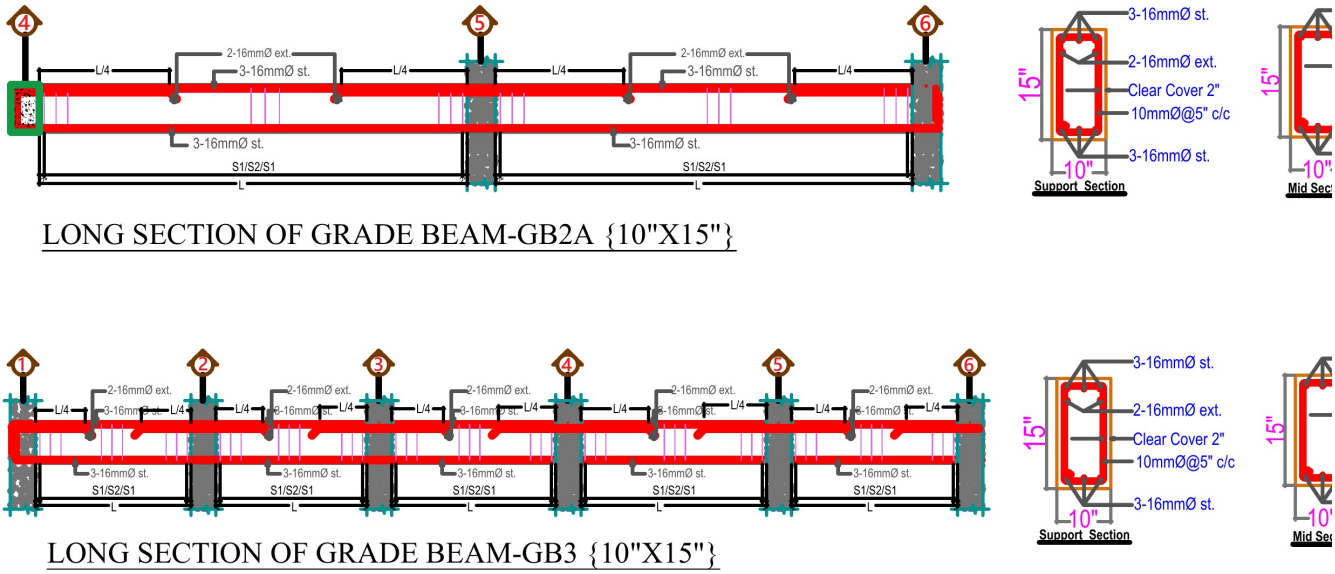
DESCRIPTION

Longitudinal sections of grade beams GB1, GB1A and GB2 with a support section and a mid section for each. They show the top and bottom bars, extra bars and stirrup spacing.

KEY INFORMATION FROM THE DRAWING

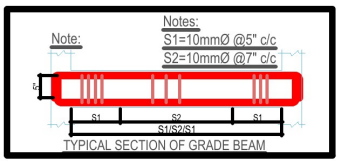
- Beam size 10" x 15", clear cover 2"
- 3-16 mm dia straight bars with 2-16 mm dia extra bars near supports (L/4)
- Stirrups 10 mm dia @ 5" c/c at supports and @ 7" c/c in the middle

Long Sections of Grade Beams GB2A and GB3, Typical Section



LONG SECTION OF GRADE BEAM-GB2A {10"X15"}

LONG SECTION OF GRADE BEAM-GB3 {10"X15"}



SECTION OF GRADE BEAM

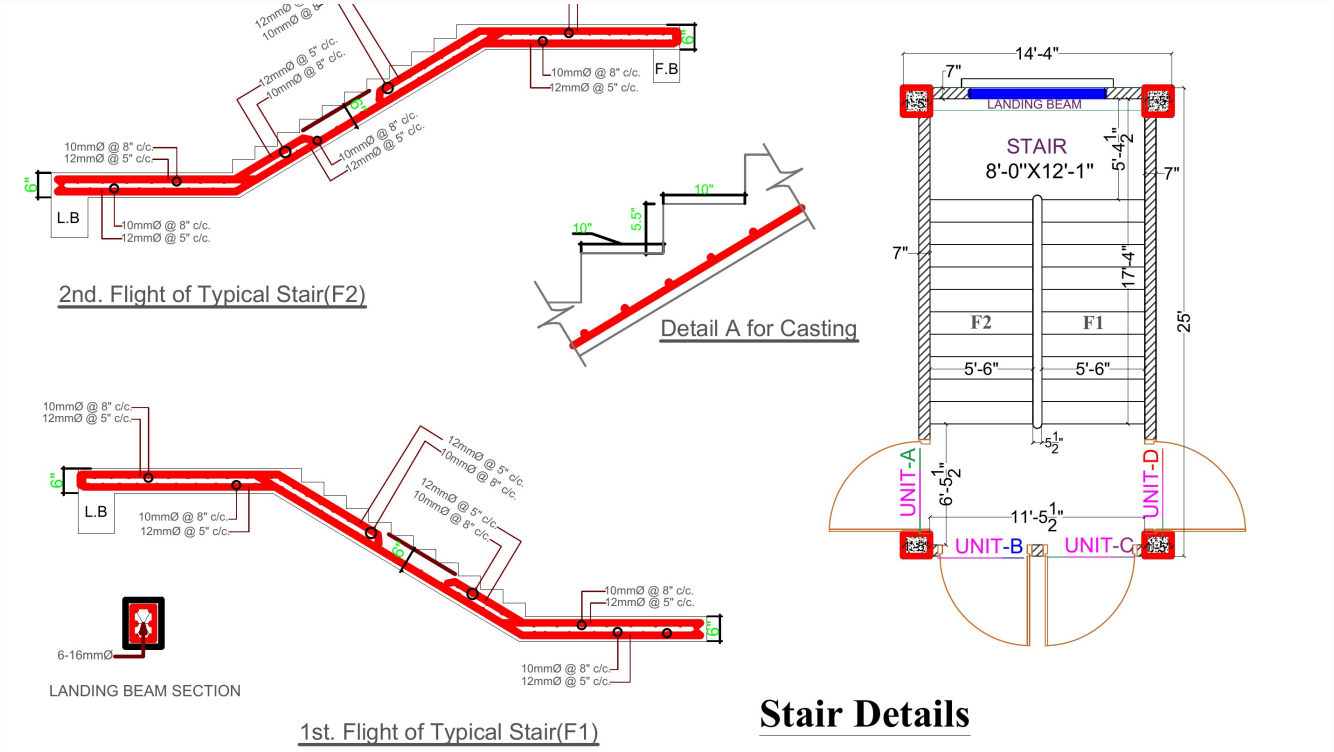
DESCRIPTION

Long sections of grade beams GB2A and GB3, with the typical grade beam cross-section and stirrup notes.

KEY INFORMATION FROM THE DRAWING

- Beam size 10" x 15", clear cover 2"
- Stirrup groups S1 = 10 mm dia @ 5" c/c and S2 = 10 mm dia @ 7" c/c
- Extra bars over supports and straight bars through the span

Stair Details



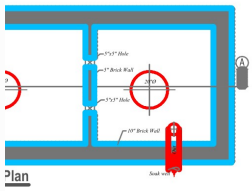
DESCRIPTION

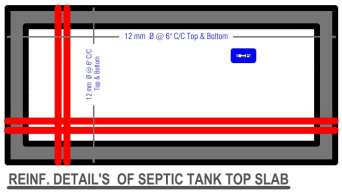
Typical stair details: the first and second flights with their reinforcement, the casting detail, the landing beam section and the stair plan with its lobby.

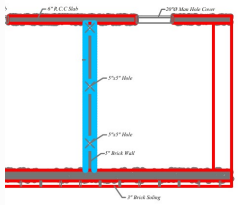
KEY INFORMATION FROM THE DRAWING

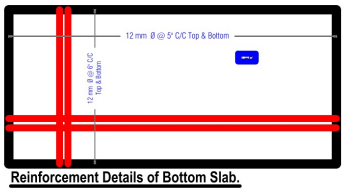
- Flights F1 and F2 with 6" risers
- Stair 8'-0" x 12'-1" with lobby 8'-0" x 4'-6"
- Main bars 12 mm dia @ 5" c/c, distribution 10 mm dia @ 8" c/c
- Landing beam with 6-16 mm dia bars

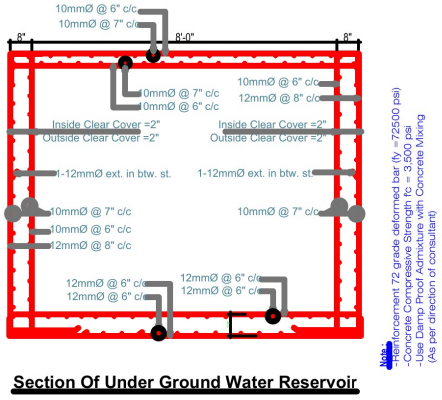
Septic Tank, Soak Well and Underground Reservoir Details

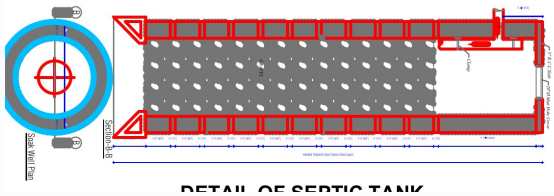












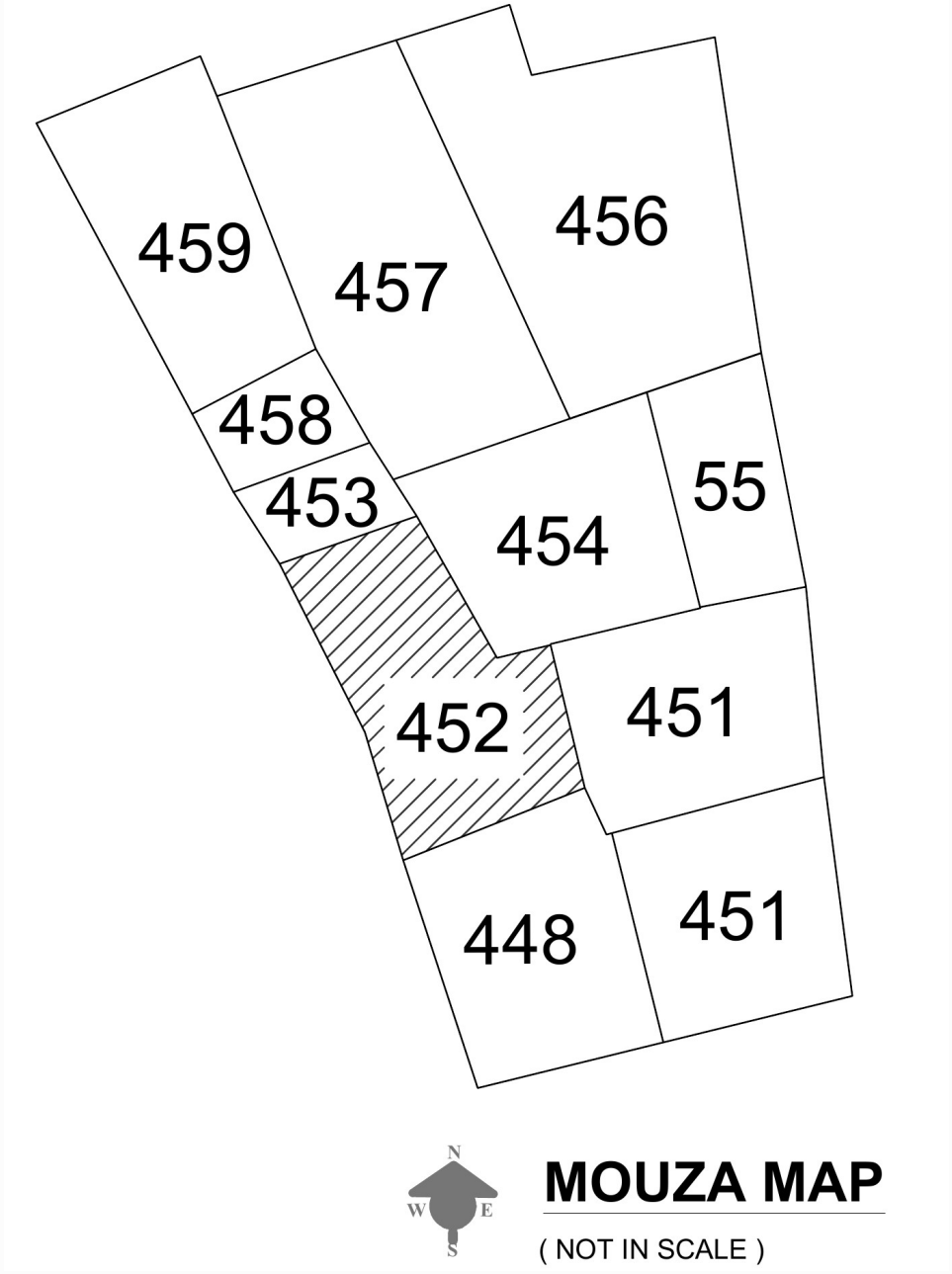
DESCRIPTION

Plans, sections and reinforcement details for the septic tank and soak well, and the section of the underground water reservoir.

KEY INFORMATION FROM THE DRAWING

- Septic tank top and bottom slabs 6" thick with 12 mm dia bars
- Brick walls 5" and 10" thick
- Reinforcement 72 grade, fy 72,500 psi; concrete f'c 3,500 psi
- Damp-proof admixture in concrete, clear cover 2"

Mouza Map



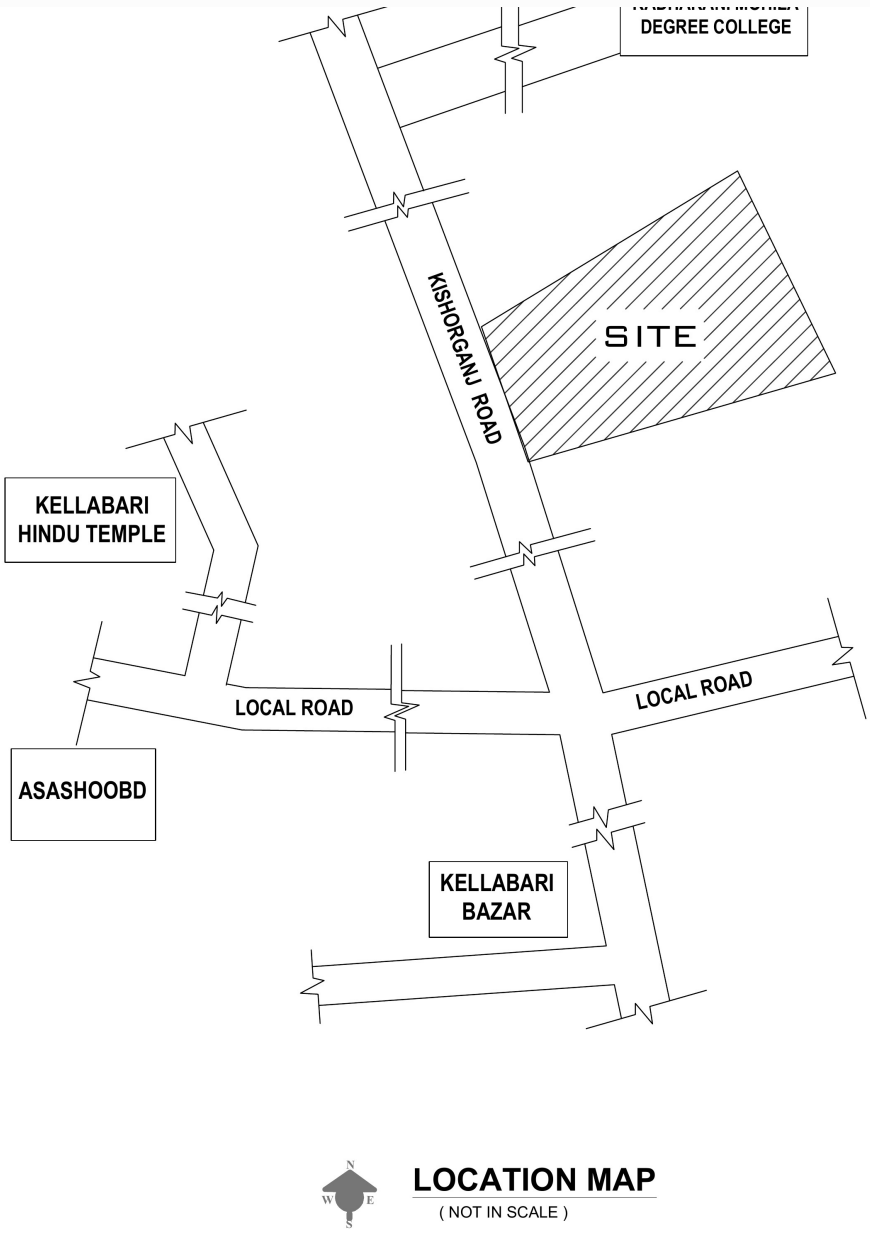
DESCRIPTION

Cadastral (mouza) map of the surrounding plots with dag numbers. The project plot, Dag 452, is hatched. Not to scale.

KEY INFORMATION FROM THE DRAWING

- Project plot Dag No. 452
- Adjoining dags 451, 453, 454, 448 and others
- Not drawn to scale

Location Map



DESCRIPTION

Sketch map showing where the site stands in relation to the nearby roads and landmarks. Not to scale.

KEY INFORMATION FROM THE DRAWING

- Radharani Mohila Degree College, Kellabari Hindu Temple and Kellabari Bazar nearby
- Kishorganj Road and local roads give access
- Not drawn to scale